Applied Linear Algebra

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https://github.com/DeviPrasad/ala2026

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What is algebra?

What is linear algebra?

What is applied linear algebra?

What is algebra?

Algebra - Wikipedia

What is linear algebra?

Linear Algebra - Wikipedia

What is applied linear algebra?

- Numerical Linear Algebra - Wikipedia
- Computational Linear Algebra - GitHub

Consider the set of linear algebraic equations:

$$
\begin{aligned}
a_{00}x_0 + a_{01}x_1 + a_{02}x_2 + \ldots + a_{0,N-1}x_{N-1} &= b_0 \\
a_{10}x_0 + a_{11}x_1 + a_{12}x_2 + \ldots + a_{1,N-1}x_{N-1} &= b_1 \\
a_{20}x_0 + a_{21}x_1 + a_{22}x_2 + \ldots + a_{2,N-1}x_{N-1} &= b_2 \\
&\ldots \\
a_{M-1,0}x_0 + a_{M-1,1}x_1 + \ldots + a_{M-1,N-1}x_{N-1} &= b_{M-1}
\end{aligned}
$$

Quiz:

1. How many equations do you see here?
2. How many unknowns do we have in this system?
3. What is each a_{ij} referred to as?

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&\ldots \\
a_{M-1,0}x_0 + a_{M-1,1}x_1 + \ldots + a_{M-1,N-1}x_{N-1} &= b_{M-1}
\end{aligned}$$

Quiz:

4. What is each x_j?
5. What do each b_m represent?

Consider the set of linear algebraic equations:

$$\begin{aligned} a_{00}x_0 + a_{01}x_1 + a_{02}x_2 + \ldots + a_{0,N-1}x_{N-1} &= b_0 \\ a_{10}x_0 + a_{11}x_1 + a_{12}x_2 + \ldots + a_{1,N-1}x_{N-1} &= b_1 \\ a_{20}x_0 + a_{21}x_1 + a_{22}x_2 + \ldots + a_{2,N-1}x_{N-1} &= b_2 \\ &\ldots \\ a_{M-1,0}x_0 + a_{M-1,1}x_1 + \ldots + a_{M-1,N-1}x_{N-1} &= b_{M-1} \end{aligned}$$

Quiz:

6. **Why do we care about such a system of equations?**

At the beginning of a semester, 55 students have signed up for ALA. The course is offered in two sections. Because of personal preferences, 20% of students in section A switch to section B, while 30% of students in section B switch to section A, resulting in a net loss of 4 students for section B.

How large are the two sections at the beginning of the semester?†

† No student dropped ALA or joined the course later.

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How large are the two sections at the beginning of the semester?

$$a + b = 55 \tag{1}$$

$$b + 0.2a - 0.3b = ?$$

$$\vdots$$

At the beginning of a semester, 55 students have signed up for ALA. The course is offered in two sections. Because of personal preferences, 20% of students in section A switch to section B, while 30% of students in section B switch to section A, resulting in a net loss of 4 students for section B.

How large are the two sections at the beginning of the semester?

$$a + b = 55 \tag{1}$$

$$b + 0.2a - 0.3b = b - 4$$

$$2a - 3b = -40 \tag{2}$$

$$\vdots$$

Solving Systems of Linear Equations

$$2x_1 + 8x_2 + 4x_3 = 2$$
$$2x_1 + 5x_2 + 1x_3 = 5$$
$$4x_1 + 10x_2 - x_3 = 1$$

```python
import numpy as np
A = np.array([[2, 8, 4],[2, 5, 1],[4, 10, -1]])
b = np.array([2, 5, 1])
x = np.linalg.solve(A, b)
assert np.allclose(np.dot(A, x), b)
assert np.equal(x, np.array([11, -4, 3])).all()
assert (x == np.array([11, -4, 3])).all()
```

Solving Systems of Linear Equations

$$
\begin{aligned}
2x_1 + 8x_2 + 4x_3 &= 2 \\
2x_1 + 5x_2 + 1x_3 &= 5 \\
4x_1 + 10x_2 - x_3 &= 1
\end{aligned}
$$

$$
\underbrace{\begin{bmatrix} 2 & 8 & 4 \\ 2 & 5 & 1 \\ 4 & 10 & -1 \end{bmatrix}}_{\mathbf{A}:\mathbb{R}^{3\times 3}}
\underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbf{x}:\mathbb{R}^{3\times 1}}
=
\underbrace{\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}}_{\mathbf{b}:\mathbb{R}^{3\times 1}}
$$

Solving Systems of Linear Equations

$$
\begin{aligned}
x_3 - x_4 - x_5 &= 4 \\
2x_1 + 4x_2 + 2x_3 + 4x_4 + 2x_5 &= 4 \\
2x_1 + 4x_2 + 3x_3 + 3x_4 + 3x_5 &= 4 \\
3x_1 + 6x_2 + 6x_3 + 3x_4 + 6x_5 &= 6
\end{aligned}
$$

```python
import numpy as np
A = np.array([[0, 0, 1, -1, -1],
              [2, 4, 2, 4, 2],
              [2, 4, 3, 3, 3],
              [3, 6, 6, 3, 6]])
b = np.array([4, 4, 4, 6])
x, _residuals, _rank, _s = np.linalg.lstsq(A, b)
assert np.allclose(np.dot(A, x), b)
```

Solving Systems of Linear Equations

Definition (Underdetermined systems ($M < N$))

- There are fewer equations than unknowns.
- Infinitely many exact solutions exist.
- so the system is underconstrained.

Definition (Overdetermined systems ($M > N$))

- There are more equations than unknowns.
- No single vector $\mathbf{x}$ satisfies $Ax = b$ exactly.
- The vector $\mathbf{b}$ simply lies outside the column space of A.